

Catheter and Guidewire Segmentation Based on CathAction Dataset

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Statement of Ethical Compliance

Data Category: No more than B

Participant Category: 0

I confirm that I have read the ethical guidelines and will follow them during this project.

1 Project Description

Endovascular interventional surgery is heavily based on the precise coordination of guidewires and catheters. Using current technologies accurately to identify and separate these devices using intraoperative X-ray images is difficult. Intraoperative X-ray images present problems such as low contrast, motion blur, and instrument overlap. Furthermore, privacy restrictions on intraoperative images cause large-scale machine learning training to be a bottleneck. Therefore, the automatic identification and segmentation of guidewires and catheters remain a challenge. Based on the latest publicly available CathAction dataset (containing both artificial models and live animal samples, with detailed annotations of guidewire/catheter categories). This study conducts deep learning research on pixel-level separation of guidewires and catheters, developing a visual recognition model with practical medical application value.

In this study, I will use intraoperative X-ray videos of vascular interventional surgeries to enable my visual model to automatically analyse guidewires, catheters, and stably segment their shapes and positions within the videos. At the beginning of the study, I will use single-frame videos or images from the artificial model portion of the CathAction dataset for supervised learning training, allowing my visual model to distinguish guidewires and catheters in static images. Next, I will fine-tune the model to improve its reliability, such as whether it maintains the same excellent recognition accuracy in blurred videos or images. Moreover, I will validate the model using the animal vascular image dataset from the CathAction dataset, testing its robustness in cross-domain scenarios from "artificial model to animal model". Finally, I will evaluate the model's recognition performance using experimental data.

The finalised model will accurately and quickly identify guidewires and catheters in vascular interventional surgery videos, providing visual recognition support for potential future high-end medical embedded surgical devices.

2 Aims & Requirements

- Separate catheters and guidewires accurately in vascular X-ray imaging.
- Achieve a processing speed comparable to normal surgical video frame rates. It still performs well in animal scenarios and has strong robustness.
- Optimise the performance of the model as much as possible to ensure that it can run on devices with lower computing power.

Table 1: Segmentation performance targets

Metric	Minimum	Target	Sprint
Overall Dice (avg. of catheter & guidewire)	≥ 0.60	≥ 0.68	≥ 0.72

Table 2: Engineering & deployment requirements

Metric	Minimum	Target	Sprint
Inference speed (FPS)	≥ 15	≥ 25	≥ 30
Single-frame latency (ms)	≤ 65	≤ 40	≤ 33
Inter-frame jitter index	≥ 0.70	≥ 0.80	≥ 0.90
Breakage rate (%)	$\leq 15\%$	$\leq 10\%$	$\leq 5\%$

Table 3: Training and Deployment Configurations

Tier	Training Scenario	Inference Scenario(512 $\times$ 512)
Minimum	512 $\times$ 512, batch 2–4, UNet-like	≥ 15 FPS
Target	512 $\times$ 512, batch 6–12, multi-model comparison	≥ 25 FPS
Sprint	Deeper/with attention, fast iteration	≥ 30 FPS

CPU ≥ 8 cores; RAM 16–32 GB; SSD ≥ 50 GB. Mixed precision (FP16/AMP) enabled by default.

3 Key Literature & Background Reading

3.1 Endovascular interventional surgery

Endovascular interventional procedures are based solely on 2D X-ray images. This makes it extremely difficult for surgeons to control catheters and guidewires within the 3D environment inside the patient’s body[1]. Guidewires and catheters, which are long and thin and have a low-contrast appearance similar to the texture of blood vessels, are easily misjudged in endovascular interventional procedures, leading to surgical risks. Surgical risks also arise from whether the surgeon immediately identifies and responds to contact/collision between the instrument tip and the vessel wall[2]. Therefore, a visual recognition and separation task for catheters and guidewires can greatly enhance surgeon feedback during surgery and reduce surgical risks[3].

3.2 Representative segmentation methods and bottlenecks

On the CathAction dataset, representative architectures such as U-Net, TransUNet, Swin-UNet, and SegViT (ViT-based) have all been applied to guidewire/catheter segmentation. Under comparable training and inference settings, Transformer-based backbones can show a small advantage on global segmentation metrics (e.g., Dice/mIoU), although the gains are limited and often come with higher computational cost[4]. However, the targets are thin, have extremely low pixel ratios, and are easily confused with blood vessel artefacts. This indicates a need to further incorporate shape/topology priors (e.g. constraints on centerline continuity and boundary clarity) into the loss function and feature modeling[5].

3.3 Computer vision models

There are corresponding learning methods for computer-aided intervention systems for various tasks[6]. In the field of deep learning, we commonly use PyTorch: the core deep learning library, including nn.Module, DataLoader, and torch.cuda.amp. These frameworks are powerful and flexible, suitable for general-purpose deep learning vision models[7]. Some framework, like TorchVision, is used for common image models and image transformations, such as basic augmentation, I/O, and pre-trained backbone functions[7]. For data augmentation, OpenCV-Python is frequently used[8]. It has powerful capabilities in reading videos/images, geometric and morphological operations, and can help analyse training datasets. Albumentations[9] and scikit-image[10] can be used as alternative data augmentation solutions. Their image augmentation functions are particularly helpful in handling segmentation mask-synchronous transformations, morphological/skeletonisation, and image metrics.

4 Development & Implementation Summary

4.1 Development environment and implementation language

PyTouch boasts a mature ecosystem for medical and interventional image segmentation, with a solid foundation in data loading, precision mixing, distributed training, and inference scripts. Furthermore, its rich and diverse open-source code libraries, built on the Python programming language, offer numerous possibilities for project development. Some of these libraries possess powerful capabilities in data processing and presentation, which facilitate research. When used with Anaconda Virtual Environment Manager, it allows code to run in a stable library environment.

4.2 Hardware development environment

The primary GPU is a single RTX 3050 Ti (4 GB), with an A100 server serving as the training server (if available). It provides powerful computing power. Ideally, model training should be performed on a corresponding cloud computing service, which offers greater flexibility. The training operating system and software can include: Ubuntu 22.04, the corresponding version of CUDA/cuDNN, a Poetry/Conda management environment, Git version control, and Weights & Biases (or MLflow) for recording experiments.

4.3 Training system composition and module division

In terms of data processing, this project adopted a unified label reading process, grayscale normalisation, and contrast stretching preprocessing strategies during the supervised learning phase.

- First, unified label reading ensures that datasets from different sources maintain consistency in category index, pixel value range (0/255 or 0/1), mask encoding (single-channel/multi-channel, RLE/PNG), and spatial dimensions. Strict isomorphic geometric transformations are performed on the image and mask (bicubic/bilinear interpolation for the image, nearest-neighbour interpolation for the mask) to avoid fragmentation or morphological shifts in elongated targets caused by interpolation.
- Second, grayscale normalisation (e.g., min-max to [0,1] or z-score) and contrast stretching (e.g., linear stretching after truncation at 1–99% quantiles, with Contrast Limited Adaptive Histogram Equalization (CLAHE) added if necessary) compress cross-domain differences in imaging intensity, enhancing the separability of weak edges and linear structures, while providing a more stable intensity baseline for subsequent morphological and topological measurements.

To address the supervision information potentially required for shape/topology-based losses (continuity constraints for slender structures), we derive centerline labels from the ground-value mask: sub-pixel-level centerline annotations are obtained through algorithms such as skeletonisation/thinning, supplemented by endpoint/branch point detection and pseudo-branch removal, resulting in a "cleaner" centerline supervision. This approach reduces the difference between small targets and background/domain distribution (extremely low foreground proportion, linear targets are easily submerged by interpolation and noise), and directly serves subsequent shape-sensitive losses (such as cLDice, topology-preserving loss, boundary/curvature consistency terms) and feature modeling (such as fine-scale branching, orientation and curvature encoding), thereby explicitly encouraging centerline connectivity, fineness, and boundary clarity during training, reducing the incidence of breaks and adhesions.

5 Data Sources

This study utilises the CathAction dataset, a large-scale, publicly available dataset designed for endovascular interventional therapy. It addresses three main tasks: guidewire/catheter segmentation, motion analysis, and tip collision detection. The dataset contains 500,000 frames of fluoroscopic sequences with motion/collision annotations. It also includes approximately 25,000 pixel-level ground truth images with separate guidewire and catheter annotations, covering 569 video segments, making it one of the most comprehensive publicly available datasets in the field in terms of both scale and task coverage[4].

Due to ethical concerns and the nature of publicly available datasets, CathAction uses two sources: adult aortic arch silicone vessel models connected to pulsatile pumps to simulate blood flow, acquired under X-ray fluoroscopy;

and animal models, where real surgery was performed on isolated animal samples, with procedures consistent with clinical practice, and imaging under contrast agents. Catheterisation procedures were performed by professionally trained endovascular interventional surgeons, and recorded using two sets of clinical fluoroscopy systems. Regarding ethics and regulations, all animal experiments and radiation-related procedures received prior ethical approval. The data was standardised and made publicly available to promote transparent and reproducible research[4].

6 Testing & Evaluation

6.1 Model evaluation metrics

After completing model training, I will calculate various evaluation metrics, including learning rate curve, confusion matrix, distortion robustness, recall, Dice accuracy, cross-validation, and other model parameters for evaluation. Simultaneously, I will assess resource and engineering capabilities, data input throughput, and peak GPU memory usage. Then, through quantitative analysis of these evaluation metrics, select the optimal machine learning model and parameters.

6.2 Core segmentation metrics (pixel level)

$$\text{Dice}_c = \frac{2TP_c}{2TP_c + FP_c + FN_c} = \frac{2|P_c \cap G_c|}{|P_c| + |G_c|}.$$

- P_c : **Predicted foreground set** — the set (mask) of all pixels predicted as class c .
- G_c : **Ground-truth foreground set** — the set of all pixels annotated as class c .
- $|\cdot|$: Set cardinality, i.e., the **number of pixels** (area in pixels).
- $P_c \cap G_c$: The set of pixels that are **overlapped** as class c by prediction and ground truth (intersection).

From these, the pixel-wise one-vs-rest confusion terms are:

- TP_c (True Positive): pixels that lie in both P_c and G_c — **predicted c and annotated c** .
- FP_c (False Positive): pixels in P_c but **not** in G_c — non- c pixels incorrectly predicted as c .
- FN_c (False Negative): pixels in G_c but **not** in P_c — c pixels missed by the prediction.
- TN_c (True Negative): pixels in neither P_c nor G_c (*do not contribute to Dice*).

The closer the Dice value is to 1, the better.

6.3 Other evaluation criteria

Specific metrics for slender structures, focusing on their topology, centerline, and connectivity. Verification of topological consistency, breakage rate (discontinuous segments along the centerline in the pred/total length), and adhesion rate (ratio of misadhered segments to blood vessels/background).

Speed and resource assessment, throughput/latency: FPS, single frame latency (ms), measured at 512×512 and target deployment resolution, respectively. Peak memory usage (batch=1), model size (ONNX/TensorRT/TS), and CPU/GPU utilisation curves are key factors in achieving clinical real-time feedback targets.

7 Project Ethics & Human Participants

This project involves data and model training that do not involve any human data privacy, and guarantees that no data privacy regulations will be violated throughout the research process. The training datasets are all publicly available and contain no human involvement or human experimentation. Ethical requirements that meet university requirements

8 BCS Project Criteria

Table 4: BCS Project Criteria

BCS Project Criteria	Illustrate
An ability to apply practical and analytical skills gained during the degree programme.	• Use the Artificial Intelligence module(COMP 219) to build the basic code architecture. • Use the database module(COMP 207) to manage model datasets. • Use the project development framework learned from COMP 208 to build code for large-scale projects. • Proficiently use knowledge of COMP 202 and COMP 281 to write code and set up compilation environments.
Innovation and/or creativity.	• The Dice+CE method, along with the introduction of three types of losses—shape, centerline, and boundary—is used to address the breakage and adhesion issues during catheter and guidewire segmentation. • The development process will take into account engineering deployment issues, ensuring it can still run on relatively low-performance computing devices.
Synthesis of information, ideas and practices to provide a quality solution together with an evaluation of that solution.	• During the development of this project, we progressed steadily by identifying, exploring, solving, and summarising problems. • We searched relevant literature and open-source resources, listed the goals to be achieved, and then combined good ideas from various parties with existing practices to select the most feasible solution. • We combined thinking and doing, using evidence to prove the feasibility of the solution.
It meets practical needs in a broad real-world context.	• This project can be deployed on embedded devices in medical institutions to assist doctors in the surgical process and help them avoid the risk of harm to patients. • In the long run, it can serve as part of a robotic automated surgery system, providing solutions for potentially more efficient treatment options in the future.
An ability to self-manage a significant piece of work.	• In this project, I will ensure timely and sufficient progress, dedicating at least 9 hours per week to project development. I will actively communicate with the supervisor to resolve any issues.
Critical self-evaluation of the process.	• Firstly, I believe I possess the motivation to solve problems and am capable of careful thought and exploration of the key aspects of the problem during the process. • However, I exhibit a negative attitude towards repetitive tasks, which leads to a loss of motivation. Therefore, this is an area I need to improve.

9 UI/UX Mockup

This image illustrates the initial model-building steps of this project. The APIs that may be used in the model code include:(torch_i=2.3, torchvision_i=0.18 segmentation-models-pytorch_i=0.3 torchmetrics_i=1.4 pytorch-lightning_i=2.3 numpy_i=1.26 scipy_i=1.11 opencv-python_i=4.10 shapely_i=2.0 networkx_i=3.3 surface-distance_i=0.1 matplotlib_i=3.9 pandas_i=2.2 hydra-core_i=1.3)

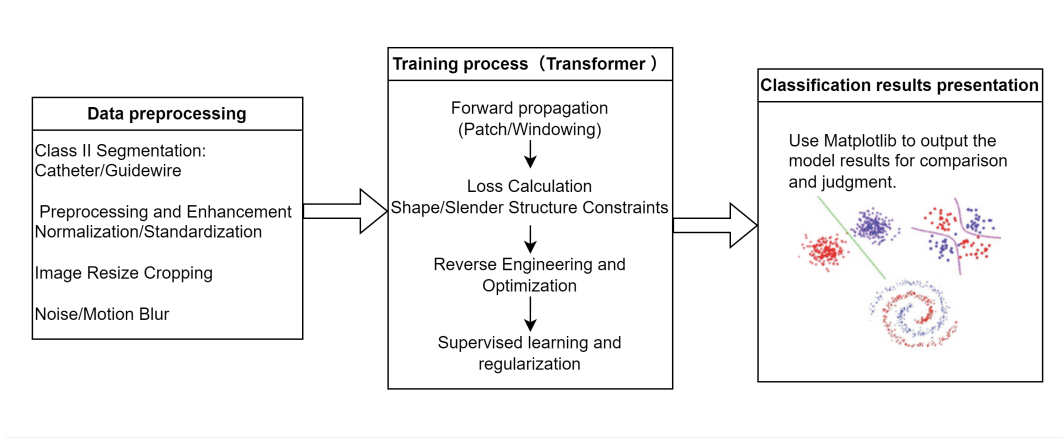


Figure 1: Project Process

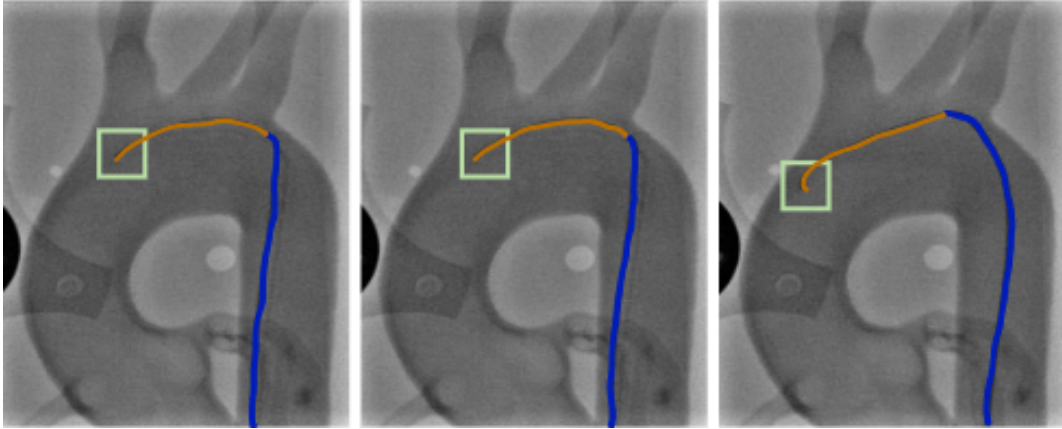
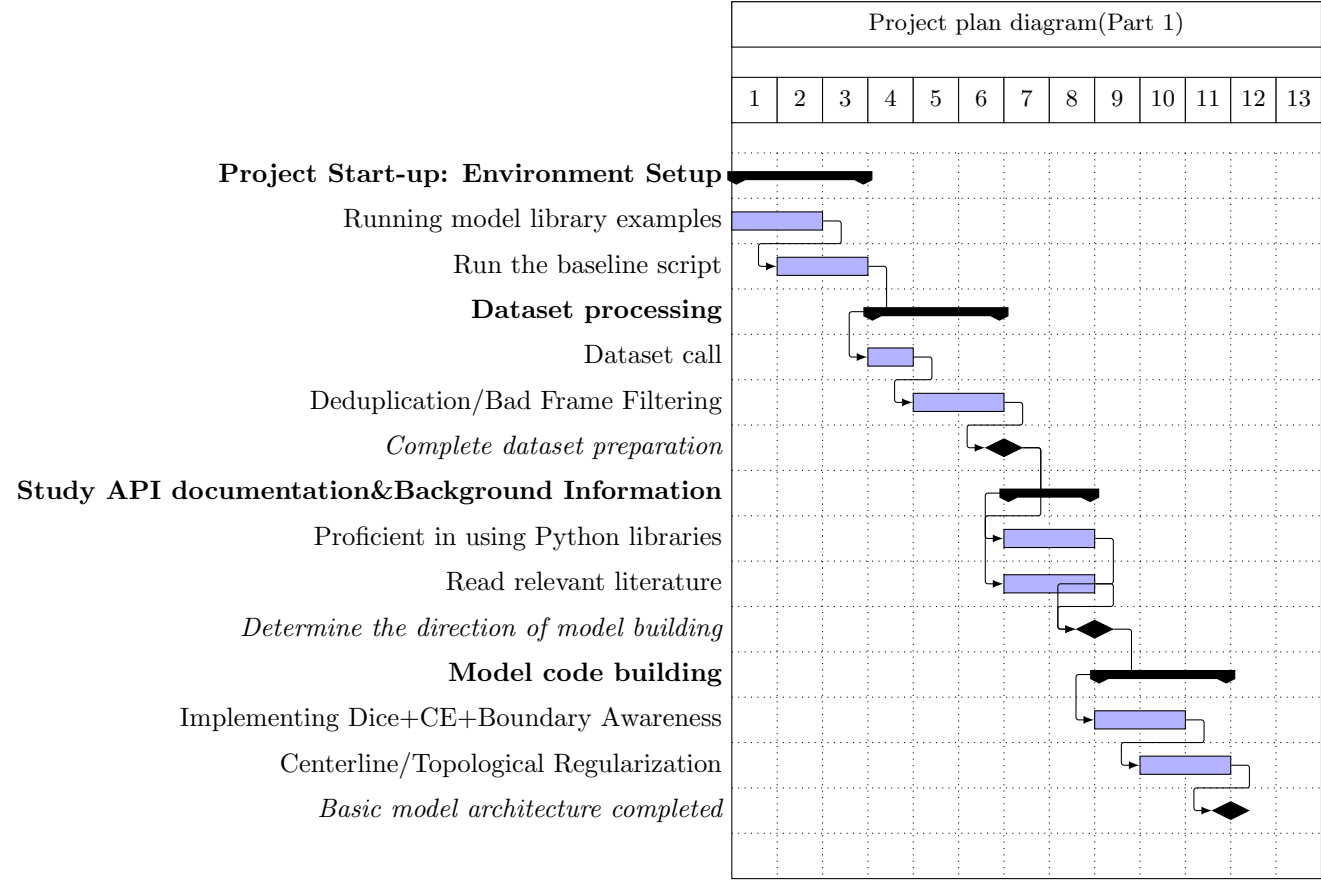
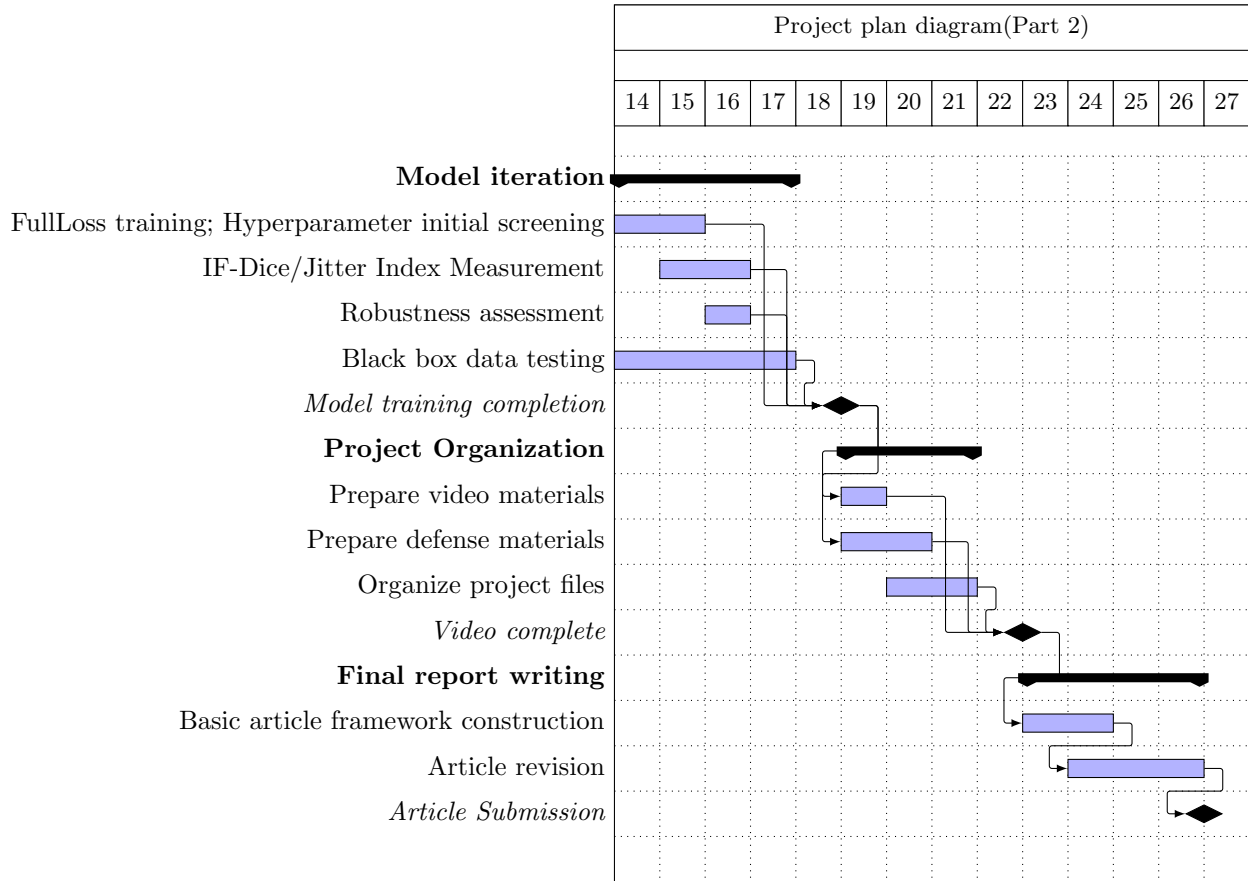


Figure 2: Segmentation result image[4]

10 Project Plan





*The time unit is weeks, starting from November.

11 Risks & Contingency Plans

Table 5: Risk assessment form

Risks	Contingencies	Likelihood	Impact
Hardware failure (GPU/SSD)	- Activate the backup computer - The training cloud server is down, immediately find other paid GPUs.	High	Delaying model training progress and losing important training parameters
Programming bugs	- Inquire about generative artificial intelligence solutions. - Contact supervisor immediately.	Low	Delayed development schedule
Dataset policy modification	Change dataset	Higt	This will severely impact the development schedule and may lead to a rewriting of the model structure.
Compute budget shortfall	- Request GPU resources from your supervisor - Try to purchase cloud GPU services	Middle	Delaying development progress

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